

Python Libraries for Data Science



MD Arshad Ahmad
15 Years+ Experience in Data Science
Mentored 100+ people

Many popular Python
toolboxes/libraries:

- NumPy

- SciPy

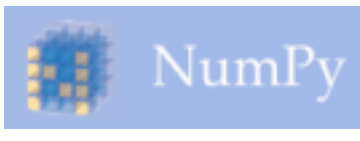
- Pandas

- SciKit-Learn

Visualization libraries

- matplotlib • Seaborn

Python Libraries for Data Science

NumPy: 

- introduces objects for multidimensional arrays and matrices, as well as functions that allow to easily perform advanced mathematical and statistical operations on those objects
- provides vectorization of mathematical operations on arrays and matrices which significantly improves the performance
- many other python libraries are built on

NumPy

Link: <http://www.numpy.org/>

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Python Libraries for Data Science

SciPy:

- collection of algorithms for linear algebra, differential equations, numerical integration, optimization, statistics and more
- part of SciPy Stack
- built on NumPy

Link: <https://www.scipy.org/scipylib/>

pandas

$$y_{it} = \beta^t x_{it} + \mu_i + \epsilon_{it}$$



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Python Libraries for Data Science

Pandas:

- adds data structures and tools
designed to work with table-like data
(similar to Series and Data Frames in R)
- provides tools for data
manipulation: reshaping, merging,
sorting, slicing, aggregation etc.
- allows handling missing data

Link: <http://pandas.pydata.org/>



Python Libraries for Data Science

SciKit-Learn:

- provides machine learning algorithms: classification, regression, clustering, model validation etc.
- built on NumPy, SciPy and matplotlib

Link: <http://scikit-learn.org/>

Python Libraries for Data Science

matplotlib:

- python 2D plotting library which produces publication quality figures in a variety of hardcopy formats
- a set of functionalities similar to those of MATLAB
- line plots, scatter plots, barcharts, histograms, pie charts etc.
- relatively low-level; some effort

needed to create advanced visualization

Link: <https://matplotlib.org/>

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Python Libraries for Data Science

Seaborn:

- based on matplotlib
- provides high level interface for

drawing attractive statistical

graphics ▪ Similar (in style) to the

popular ggplot2 library in R

Link: <https://seaborn.pydata.org/>

